DATA STRUCTURES

LAB PROGRAM

7. Write a C program to implement Array operations such as Insert, Delete and display

```
#include <stdio.h>


int main()

{

   int arr[100], n, i, pos, value;


   printf("Enter number of elements: ");

   scanf("%d", &n);


   printf("Enter array elements:\n");

   for(i = 0; i < n; i++)

      scanf("%d", &arr[i]);


   // Insert Operation

   printf("Enter position to insert: ");

   scanf("%d", &pos);


   printf("Enter value to insert: ");

   scanf("%d", &value);


   for(i = n; i >= pos; i--)

      arr[i] = arr[i - 1];
```

```
arr[pos - 1] = value;
n++;

printf("Array after insertion:\n");
for(i = 0; i < n; i++)
    printf("%d ", arr[i]);

// Delete Operation
printf("\n\nEnter position to delete: ");
scanf("%d", &pos);

for(i = pos - 1; i < n - 1; i++)
    arr[i] = arr[i + 1];

n--;

printf("Array after deletion:\n");
for(i = 0; i < n; i++)
    printf("%d ", arr[i]);

// Display Operation
printf("\n\nArray elements are:\n");
for(i = 0; i < n; i++)
    printf("%d ", arr[i]);
```

```
    return 0;
}
```

Output

```
Enter number of elements: 5
Enter array elements:
10 20 70 40 50
Enter position to insert: 3
Enter value to insert: 30
Array after insertion:
10 20 30 70 40 50

Enter position to delete: 4
Array after deletion:
10 20 30 40 50

Array elements are:
10 20 30 40 50

=== Code Execution Successful ===
```